Isentropic Flow Analysis

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1 Core Concept

This concept of isentropic flow is mainly used in fluid mechanics and thermodynamics, mainly in the high-speed aerodynamics area. Here the flow is adiabatic and reversible. For a flow to be truly isentropic, it must satisfy these two conditions:

- **Adiabatic** ($\delta q = 0$) — There is no heat transfer between the fluid and its surroundings. The fluid does not gain heat from a nearby combustion source, nor does it lose heat to the surroundings.
- **Reversible** — There are no friction losses within the flow. This means there is no internal friction (viscosity) and no turbulence.

If both are true, then the change in entropy is zero:

$$ds = 0 \Rightarrow s = \text{constant}$$

Stagnation Conditions: Stagnation conditions (T_0, P_0, ρ_0) represent the temperature, pressure, and density the flow would have if brought to rest isentropically. They serve as reference conditions from which all flow properties are normalized. At $M = 0$, static conditions equal stagnation conditions. As Mach number increases, static temperature, pressure, and density all fall below their stagnation values as thermal energy converts to kinetic energy.

What This Project Does: This project calculates and visualizes the four fundamental isentropic flow relations — temperature ratio T/T_0, pressure ratio P/P_0, density ratio ρ/ρ_0, and area ratio A/A^* — as functions of Mach number from $M = 0$ to $M = 5$. It also provides a lookup table for any given Mach number, serving as a computational version of the isentropic flow tables used in aerospace engineering practice.

2 Governing Equations

1 — Temperature Ratio:

$$\frac{T}{T_0} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-1}$$

2 — Pressure Ratio:

$$\frac{P}{P_0} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-\frac{\gamma}{\gamma-1}}$$

3 — Density Ratio:

$$\frac{\rho}{\rho_0} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-\frac{1}{\gamma-1}}$$

4 — Area Ratio:

$$\frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma - 1}{2} M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

These are the same equations used in Project 2 — Rocket Nozzle Flow. The difference is that here they are plotted across the full Mach number range rather than evaluated at a single exit point.

3 Key Results

Mach Number (M)	T/T_0	P/P_0	ρ/ρ_0	A/A^*
0.01	1.0000	0.9999	1.0000	57.8738
0.50	0.9524	0.8430	0.8852	1.3398
1.00	0.8333	0.5283	0.6339	1.0000
2.00	0.5556	0.1278	0.2300	1.6875
5.00	0.1667	0.0019	0.0113	25.0000

4 Physical Insights

Connection to Project 2: These equations are the same ones used in the Rocket Nozzle Flow project. The difference is that instead of evaluating them at one specific exit point, here they are plotted across all Mach numbers to show how each property changes continuously with flow speed. Seeing the full curve reveals behavior that a single-point calculation cannot show.

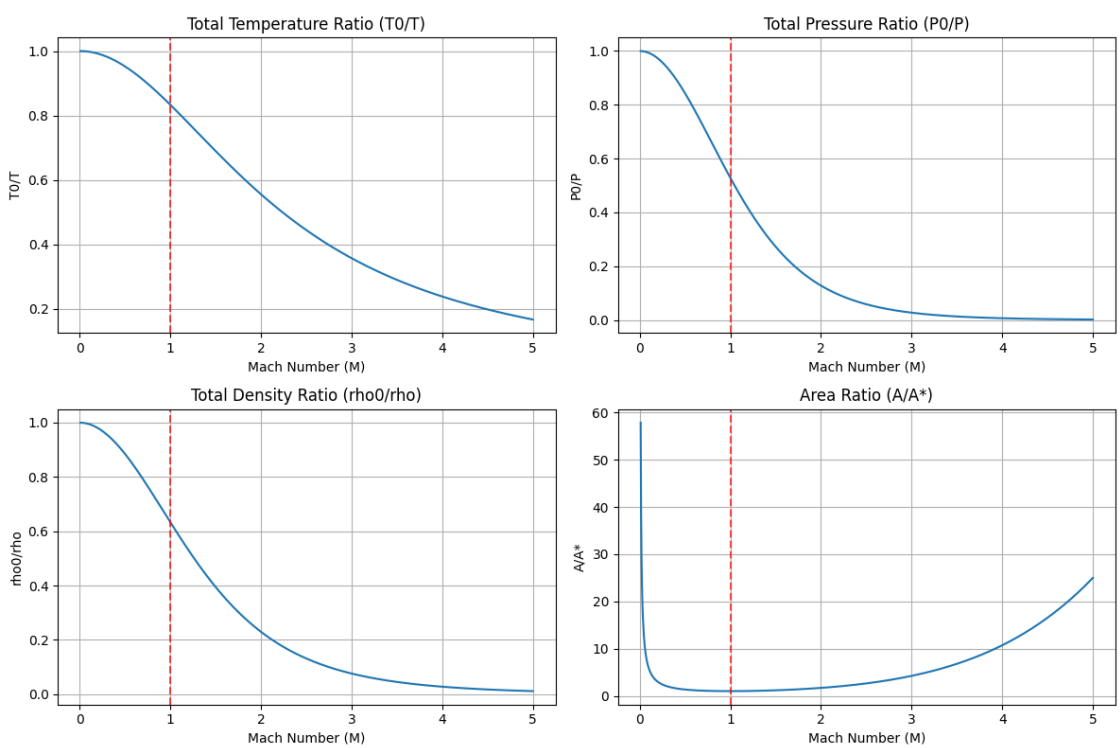
Pressure drops faster than temperature: In the plots, P/P_0 drops much more steeply than T/T_0. The reason is the exponent — pressure is raised to $\gamma/(\gamma - 1) = 3.5$ for air, while the temperature exponent is 1. This has real consequences for aerospace design. At high Mach numbers, pressure drops drastically compared to temperature. In reverse, when air hits an aircraft and slows

to stagnation, pressure spikes far more dramatically than temperature. This is why aerodynamic pressure loads on aircraft structures are often the dominant design concern at high speeds, not just thermal loads.

The A/A^* curve — two branches: The area ratio curve has two branches that meet at $M = 1$, where $A/A^* = 1$ — the minimum-area point, the throat. The left branch represents subsonic flow in the converging section of a nozzle, while the right branch represents supersonic flow in the diverging section. This is the mathematical foundation of converging-diverging nozzle design from Project 2, now visible as a complete curve rather than a single calculated point.

5 Assumptions Made

- Calorically perfect gas — constant specific heats
- Steady, one-dimensional flow
- No heat transfer — adiabatic
- No friction or viscous effects
- No normal shocks



Isentropic Flow Analysis Plots